

Nathan Wong Yong Jie

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EDUCATION

National University of Singapore

Bachelor of Computing in Computer Science (CGPA: 4.35/5.0)

Graduation: May 2028

Relevant Coursework:

- Object-Oriented Programming, Data Structures & Algorithm, AI & Machine Learning
- Computer Organization, Operating Systems, Software Engineering
- Linear Algebra, Multivariable Calculus, Statistics & Probability

PROJECTS

Spam Message Detector

Mar 2026

Python, Pandas, NumPy, NLTK, Scikit-learn, Matplotlib, Seaborn

- Built an **NLP-based binary classifier** to detect spam messages, achieving **97.8%** accuracy and **0.91** F1 score using SVM on a dataset of 5,572 SMS messages
- Implemented a full ML pipeline: text preprocessing (stopword removal, Porter stemming), TF-IDF vectorization with unigram/bigram features, and stratified train-test splitting
- Trained and compared 5 classification models (SVM, Naive Bayes, Logistic Regression, Random Forest, Decision Tree), evaluating each on accuracy, precision, recall, and F1 score

SnapMap

May 2025

Expo, TypeScript, JavaScript, React, React Native, Tailwind CSS, Supabase

- Engineered a cross-platform mobile app enabling users to pin and revisit geo-tagged memories on an interactive map, implementing custom **React Native Maps markers** and **callouts** to support rich location-based content browsing.
- Architected and deployed a serverless backend using Supabase, implementing RESTful APIs for secure user authentication and PostgreSQL data persistence.
- Integrated Geoapify API to render lightweight static map thumbnails for memory previews, reducing rendering overhead compared to loading full interactive maps in list views.

VitTv

Jan 2026

JavaScript, React, Tailwind CSS, Vite

- Built a **retro TV-themed web app** featuring curated video playback and custom UI controls, leveraging **Vite's** fast HMR and optimized bundling to achieve sub-second load times and a smooth browsing experience.
- Designed a fully responsive UI with custom **Tailwind CSS** utility classes, structuring styles for component reusability and ensuring consistent rendering across desktop and mobile viewports.

EXTRACURRICULAR ACTIVITY

Raffles Hall Developers

Aug 2025 - Present

- Collaborated with other hall members to build a project together using Git and Docker.
- Orchestrated a medium-scale inter-hall hackathon, coordinating logistics and technical requirements for 40+ participants across multiple residential communities.

SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java

Frameworks & Tools: React, React Native, Expo, Tailwind CSS, Scikit-Learn, TensorFlow, PyTorch

Database: MySQL, PostgreSQL, Supabase

Tools: Git, Docker